

Maximum and Minimum Values

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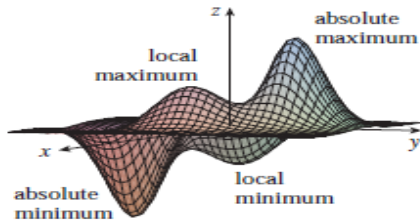
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Consider a function $z = f(x, y)$.

Definition f has a **local maximum** at (a, b) if $f(x, y) \leq f(a, b)$ when (x, y) is near (a, b) . Similarly, f has a **local minimum** at (a, b) if $f(x, y) \geq f(a, b)$ when (x, y) is near (a, b) .

If $f(x, y) \leq f(a, b)$ for all values of $(x, y) \in \text{Dom}(f)$, then f has an **absolute maximum** at (a, b) . Similarly, if $f(x, y) \geq f(a, b)$ for all values of $(x, y) \in \text{Dom}(f)$, then f has an **absolute minimum** at (a, b) .



Suppose that the partial derivatives f_x and f_y exist for $f(x, y)$.

Theorem If f has a local maximum or minimum at a point (a, b) , then $f_x(a, b) = 0$ and $f_y(a, b) = 0$.

Proof

Definition A point (a, b) is called a **critical point** (or a **stationary point**) of f if $f_x(a, b) = 0$ and $f_y(a, b) = 0$, or if one of these partial derivatives does not exist.

Example 1. Find the critical points of $f(x, y) = e^{9x-x^2+y^2}$

In order to classify critical points, we use the second derivative test.

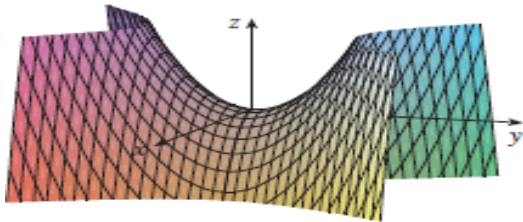
Theorem Suppose the second partial derivatives of f are continuous around (a, b) and $f_x(a, b) = f_y(a, b) = 0$ (i.e. (a, b) is a critical point). Define

$$D = \Delta = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix} = f_{xx}f_{yy} - [f_{xy}]^2$$

- If $D(a, b) > 0$ and $f_{xx}(a, b) > 0$, then (a, b) is a local minimum
- If $D(a, b) > 0$ and $f_{xx}(a, b) < 0$, then (a, b) is a local maximum
- If $D(a, b) < 0$, then (a, b) is neither a local maximum nor a minimum

Remark If $D = 0$, then the test does not work: (a, b) might be a local max, a local min or a **saddle point**.

A critical point that is neither a local min nor a local max is called a saddle point.



Proof of the Second Derivative Test

Let $f(x, y)$ is a 'nice function', and $f_x(a, b) = f_y(a, b) = 0$.
Then we have

$$\begin{aligned} f(a+h, b+k) &\sim f(a, b) + f_x(a, b)h + f_y(a, b)k \\ &\quad + \frac{1}{2} (f_{xx}(a, b)h^2 + 2f_{xy}(a, b)hk + f_{yy}(a, b)k^2) + E(h, k) \end{aligned} \quad (1)$$

where $E(h, k)$ is the **error term**.

Since $f_x(a, b) = f_y(a, b) = 0$, equation (1) becomes

$$f(a+h, b+k) \approx f(a, b) + \frac{1}{2} \underbrace{(f_{xx}(a, b)h^2 + 2f_{xy}(a, b)hk + f_{yy}(a, b)k^2)}_{Q(h, k)}$$

The discriminant of $Q(h, k)$ is

$$D = f_{xx}(a, b)f_{yy}(a, b) - f_{xy}(a, b)^2$$

Claim Consider a quadratic form

$$Q(h, k) = ah^2 + 2bhk + ck^2$$

with the discriminant $D = ac - b^2$

- ▶ If $D > 0$ and $a > 0$ then $Q(h, k) > 0$ for $(h, k) \neq (0, 0)$
- ▶ If $D > 0$ and $a < 0$ then $Q(h, k) < 0$ for $(h, k) \neq (0, 0)$
- ▶ If $D < 0$ then $Q(h, k)$ takes both positive and negative values.

Proof of the Claim

$$\begin{aligned} Q(h, k) &= ah^2 + 2bhk + ck^2 = a\left(h + \frac{bk}{a}\right)^2 + \left(c - \frac{b^2}{a}\right)k^2 \\ &= a\left(h + \frac{bk}{a}\right)^2 + \frac{D}{a}k^2 \end{aligned}$$

Now, using the claim, we have

► If $D > 0$ and $f_{xx}(a, b) > 0$ then $Q(h, k) \geq 0$. Thus, $f(a + h, b + k) - f(a, b) > 0$ when h and k are small. Hence, $f(a, b)$ is a local min.

► If $D > 0$ and $f_{xx}(a, b) < 0$ then $Q(h, k) \leq 0$. Thus,

$f(a + h, b + k) - f(a, b) < 0$ when h and k are small. Hence, $f(a, b)$ is a local max.

► If $D < 0$ then $Q(h, k)$ takes both positive and negative values. Thus, (a, b) is neither a max nor a min.

The proof of the second derivative test optional.

Now, we can solve some examples:

Example 2. For the following function

- Find all critical points
- Classify the critical points (i.e. determine whether they are local min, local max, or saddle points).

▶ $f(x, y) = 9 - 2x + 4y - x^2 - 4y^2$

Example 2 cont'd For the following function

- Find and classify all critical points

▶ $f(x, y) = x^4 + y^4 - 4xy + 2$

Example 2 cont'd For the following function

- Find and classify all critical points

▶ $f(x, y) = 2x^3 + xy^2 + 5x^2 + y^2$

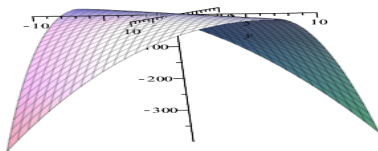
Exercise 3. For the following functions

- Find and classify all critical points of

- ▶ $f(x, y) = x^3y + 12x^2 - 8y$

- ▶ $f(x, y) = (2x - x^2)(2y - y^2)$

Remark $f(x, y) = 1 + 2xy - x^2 - y^2$



Example Find and classify critical points of $f(x, y) = x^6 - y^8$.

Solution. As $f_x = 6x^5$ and $f_y = -8y^7$, the only critical point is $(0, 0)$. However

$$\Delta = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix} = \begin{vmatrix} 30x^4 & 0 \\ 0 & -56y^6 \end{vmatrix}$$

Clearly, $\Delta(0, 0) = 0$, hence we cannot use the second derivative test. Note that. $f(0, 0) = 0$.

If we approach $(0, 0)$ along the x -axis, we have $f(x, 0) = x^6 > 0$, hence $f(0, 0)$ cannot be a local max.

If we approach $(0, 0)$ along the y -axis, we have $f(0, y) = -y^8 < 0$, hence $f(0, 0)$ cannot be a local min.

Therefore, $(0, 0)$ is a saddle point.

Recall a classical result in Calculus I: If $y = f(x)$ is continuous on a closed interval $[a, b]$, then $f(x)$ attains its absolute maximum and minimum on $[a, b]$.

We have a variant of this theorem in Calculus III:

Theorem If f is continuous on a closed, bounded set D of $\mathbb{R}^2$, then f attains its absolute maximum and absolute minimum on D .

The Technique to find the Extreme Values

- Find all critical points of f in D
- Evaluate f at the critical points (found in previous step)
- Find the extreme values of f on the boundary of D
- Compare the values of f obtained above.

Example 4. Find the absolute maximum and the absolute minimum values of the function $f(x, y) = 2x^3 + 2xy^2 - x - y^2$ over the unit disk $x^2 + y^2 \leq 1$.

Solution. First, let us find the critical points of $f(x, y)$ inside the given region. Then we must have

$$f_x = 6x^2 + 2y^2 - 1 = 0 \quad (2)$$

$$f_y = 4xy - 2y = 2y(2x - 1) = 0 \quad (3)$$

Clearly, equation (3) implies that $y = 0$ or $x = \frac{1}{2}$. Now, if $y = 0$, then equation (3) gives us $x = \mp \frac{1}{\sqrt{6}}$. However, if $x = \frac{1}{2}$, then equation (3) turns into $2y^2 = -\frac{1}{2}$ which has no real solutions. Hence, the only critical points are $P_1(\frac{1}{\sqrt{6}}, 0)$ and $P_2(-\frac{1}{\sqrt{6}}, 0)$. Now, let us examine the boundary of the disk.

Clearly, $x^2 + y^2 = 1 \implies y = \pm\sqrt{1-x^2}$. Now,

$$h(x) := f(x, \pm\sqrt{1-x^2}) = x^2 + x - 1, \quad -1 \leq x \leq 1$$

Since $h'(x) = 2x + 1 = 0$ implies $x = -\frac{1}{2}$, we obtain the following points on the boundary

$$P_3(-\frac{1}{2}, \frac{\sqrt{3}}{2}) \text{ and } P_4(-\frac{1}{2}, -\frac{\sqrt{3}}{2}).$$

Finally, we must include the end points $x = 1$ and $x = -1$ which give us the points

$$P_5(1, 0) \text{ and } P_6(0, 1).$$

Now, we test the points we found on the function $f(x, y)$:

$$f(P_1) = -\frac{1}{3}\sqrt{\frac{2}{3}}$$

$$f(P_2) = \frac{1}{3}\sqrt{\frac{2}{3}}$$

$$f(P_3) = f(P_4) = -\frac{5}{4} \quad (\text{absolute minimum value})$$

$$f(P_5) = 1 \quad (\text{absolute maximum value})$$

$$f(P_6) = -1$$

Example 5. Find the extreme values (i.e. absolute maximum and absolute minimum) values of $f(x, y) = 3 + xy - x - 2y$ on D , where D is a closed triangular region with vertices $(1, 0)$, $(5, 0)$ and $(1, 4)$.

Example 6. Find the extreme values of $f(x, y) = 4x + 6y - x^2 - y^2$ on the region $D = \{(x, y) : 0 \leq x \leq 4, 0 \leq y \leq 5\}$.

Example 7. Find the extreme values of $f(x, y) = 2x^3 + y^4$ on the region $D = \{(x, y) : x^2 + y^2 \leq 1\}$.

Example 8. Find the values of a and b so that

$$\int_a^b (6 - x - x^2) dx$$

has its largest possible value.

Solution. Let us define a function from the given integral via

$$F(a, b) = \int_a^b (6 - x - x^2) dx \text{ where } a < b.$$

The boundary of the domain of F is the line $b = a$ in the ab -plane, and clearly $F(a, a) = 0$.

Now we check the interior critical points:

$$\frac{\partial F}{\partial a} = -(6 - a - a^2) = 0 \implies a = -3, 2$$

$$\frac{\partial F}{\partial b} = 6 - b - b^2 = 0 \implies b = -3, 2.$$

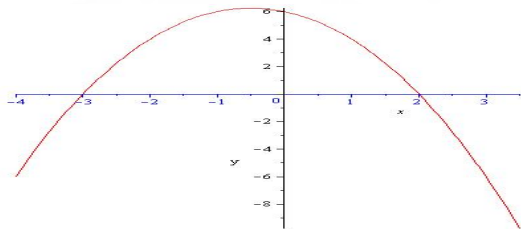
Since $a \leq b$, there is only one interior critical point namely $(-3, 2)$.

Note that

$$F(-3, 2) = \int_{-3}^2 (6 - x - x^2) dx = \frac{125}{6}.$$

Therefore $a = -3$ and $b = 2$.

Notice that $F(-3, 2)$ gives the area under the parabola $y = 6 - x - x^2$ that is above the x -axis.



Example 9. Suppose that a circular plate in the shape of $x^2 + y^2 \leq 1$ is heated so that the temperature at any point is given By

$$T(x, y) = x^2 + 2y^2 - x.$$

Find the **hottest** and **coolest** points on this plate.

Solution. First, we find the critical points of $T(x, y)$:

$$T_x(x, y) = 2x - 1 = 0, \quad T_y(x, y) = 4y = 0$$

$$\implies x = \frac{1}{2}, y = 0 \implies \left(\frac{1}{2}, 0\right) \text{ is the only critical point.}$$

$$T\left(\frac{1}{2}, 0\right) = -\frac{1}{4}.$$

On the boundary $x^2 + y^2 = 1$, we have $y = \pm\sqrt{1 - x^2}$, and $T(x, y)$ becomes

$$\phi(x) = T(x, \pm\sqrt{1 - x^2}) = -x^2 - x + 2, \quad -1 \leq x \leq 1.$$

$$\frac{d\phi}{dx} = -2x - 1 = 0 \implies x = -\frac{1}{2}, y = \pm \frac{\sqrt{3}}{2}.$$

$$T\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right) = \frac{9}{4}, \quad T\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right) = \frac{9}{4},$$

$$T(-1, 0) = 2, \quad T(1, 0) = 0.$$

Hence, the **hottest points** are $\left(-\frac{1}{2}, \pm \frac{\sqrt{3}}{2}\right)$ with the temperature $\frac{9}{4}^{\circ}$.

Similarly, the **coldest point** is $\left(\frac{1}{2}, 0\right)$ with the temperature $-\frac{1}{4}^{\circ}$.